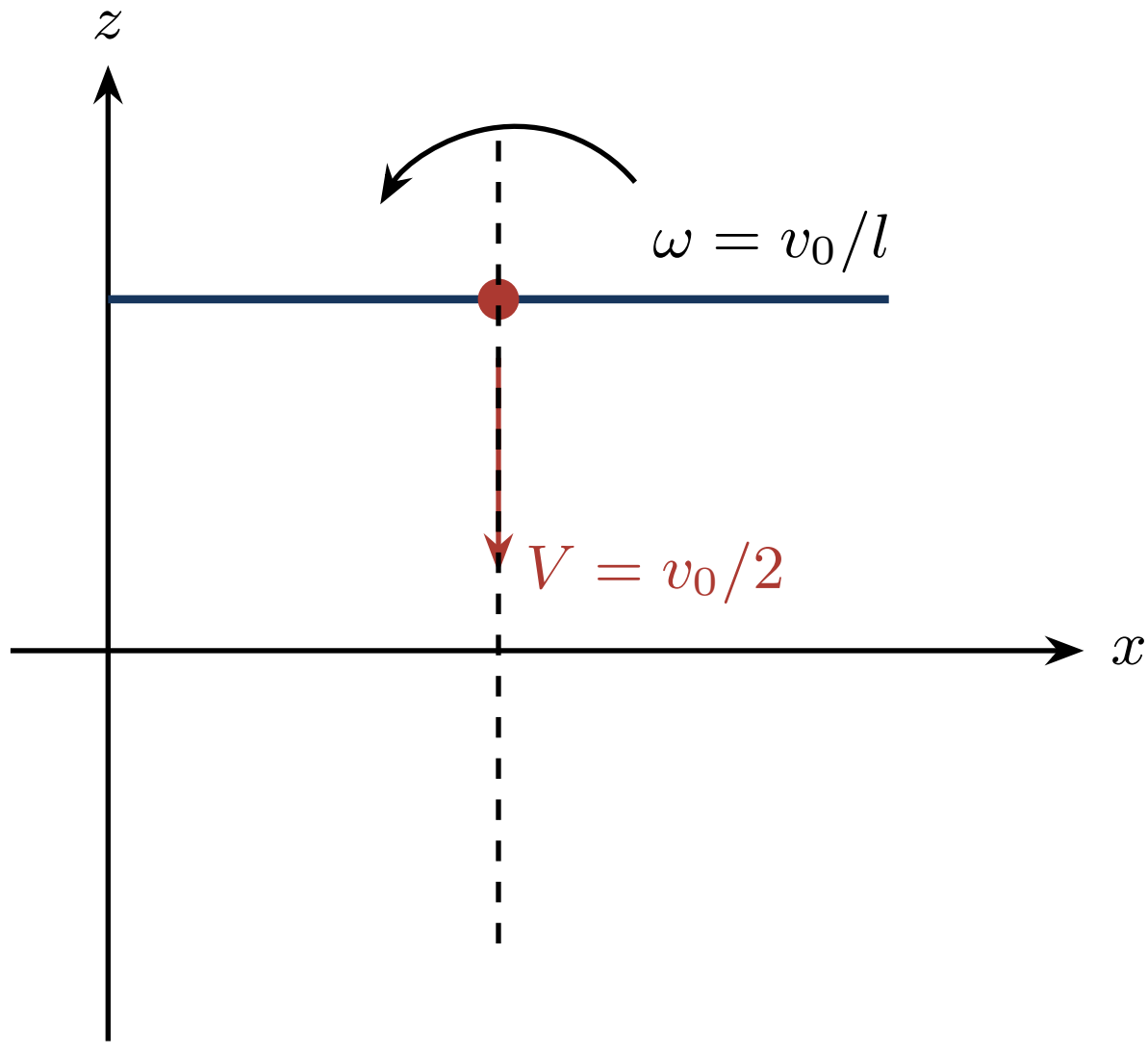
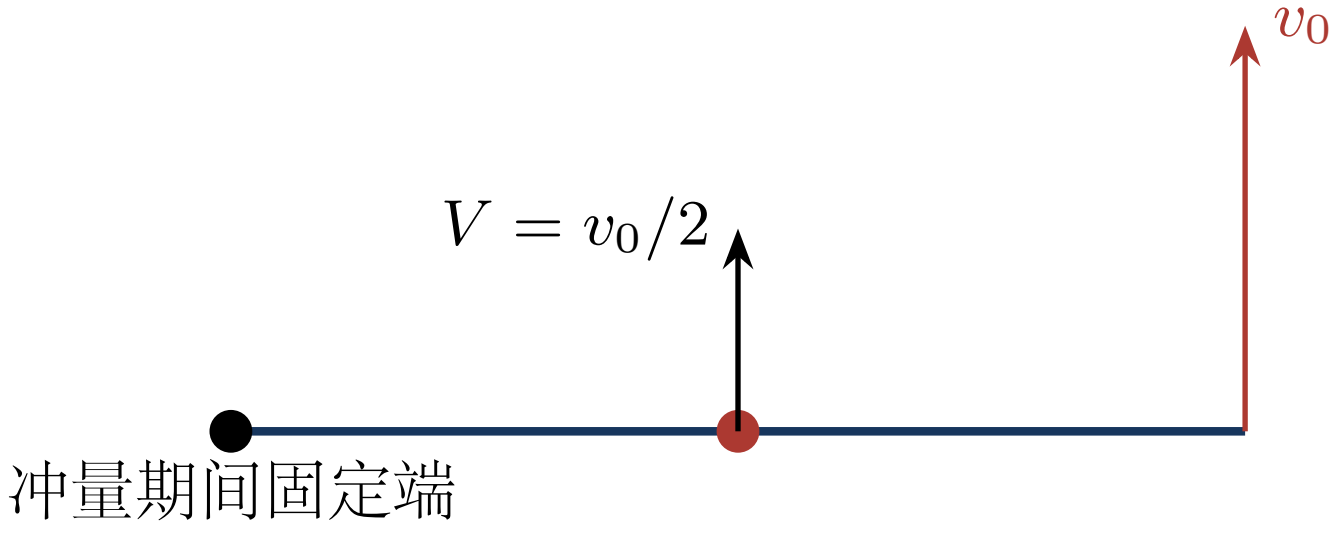
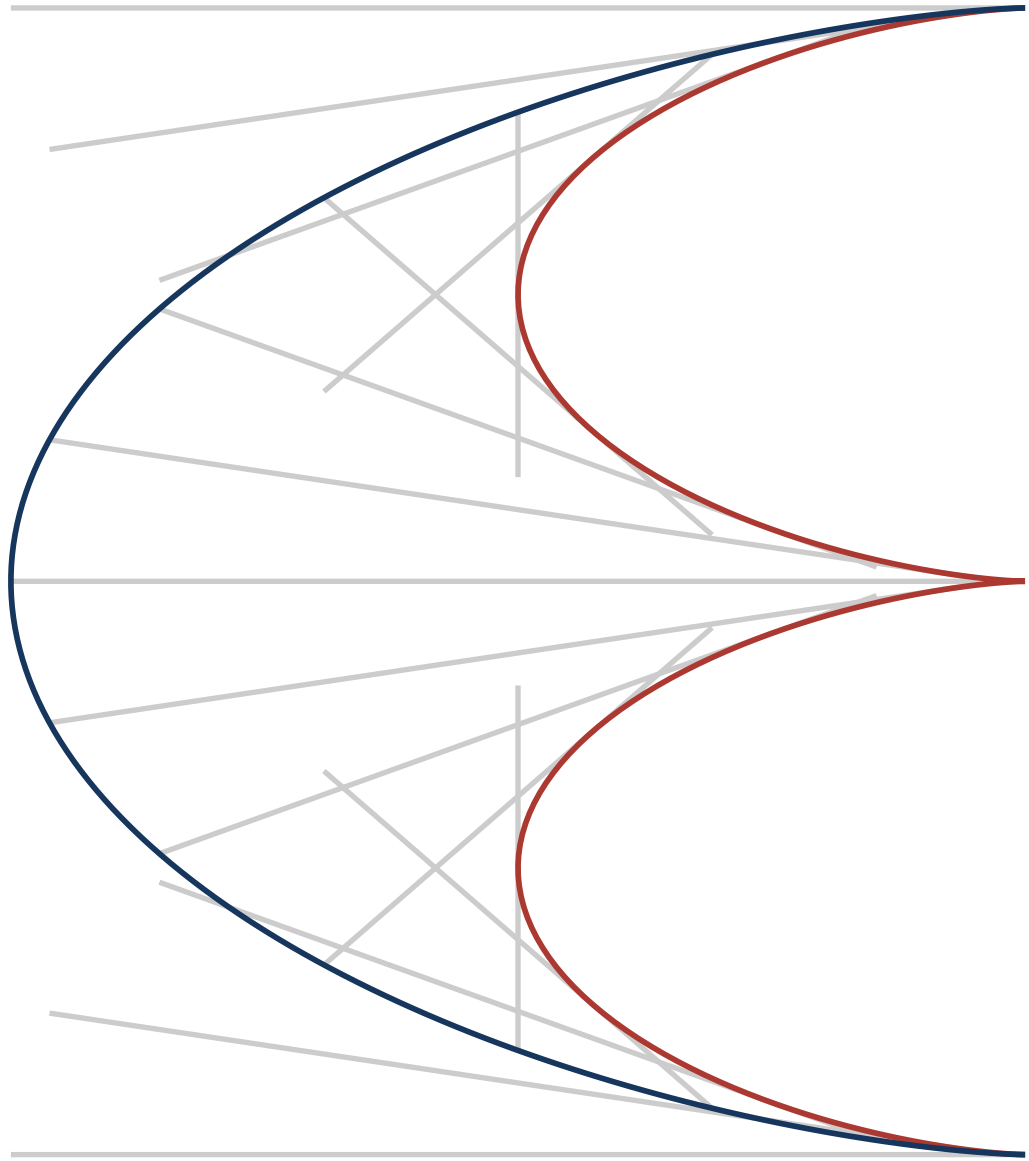


局部下落：质心匀速向下，笔绕质心逆时针转动

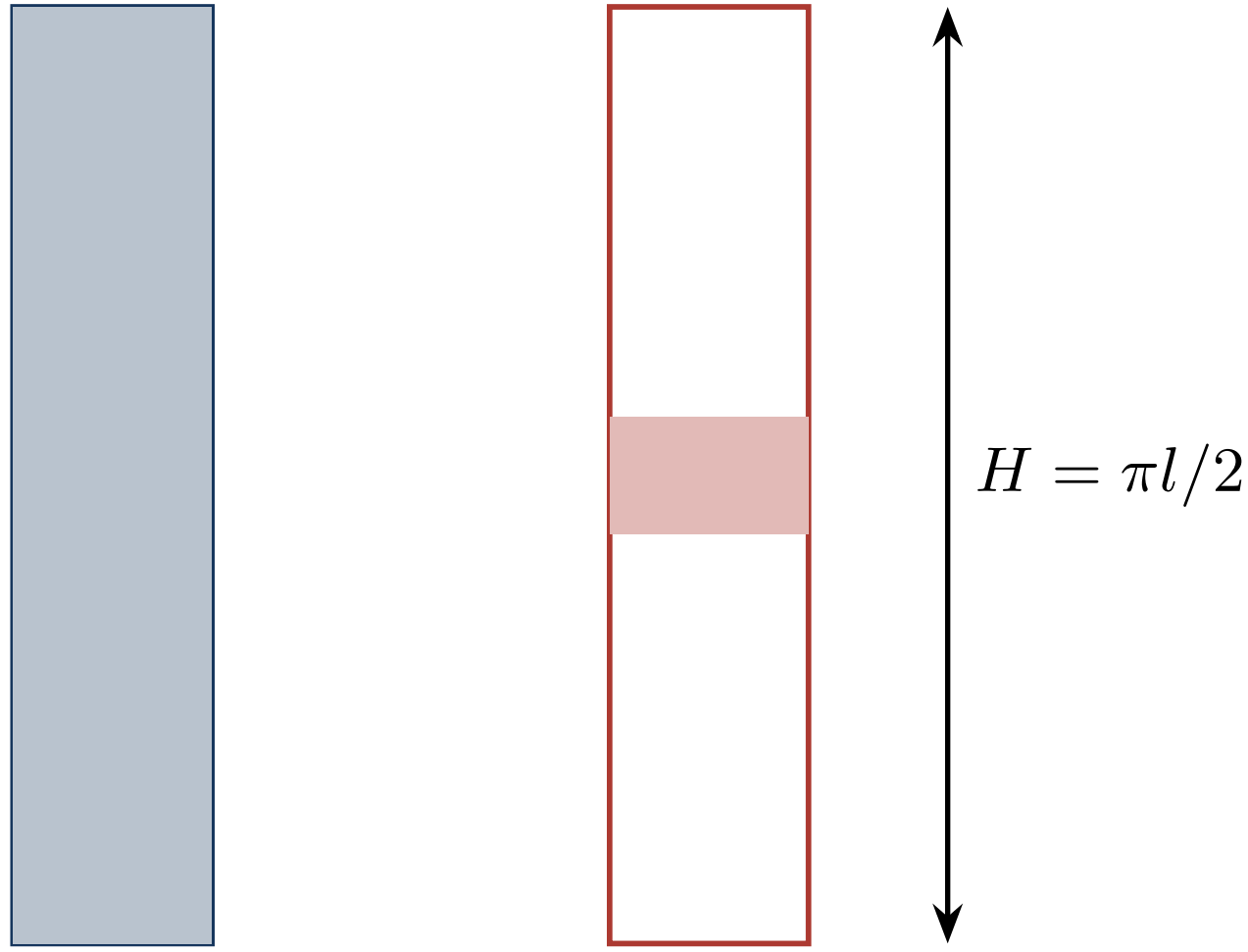


释放后 $I_C = ml^2/12$；平动与转动动能之和保持

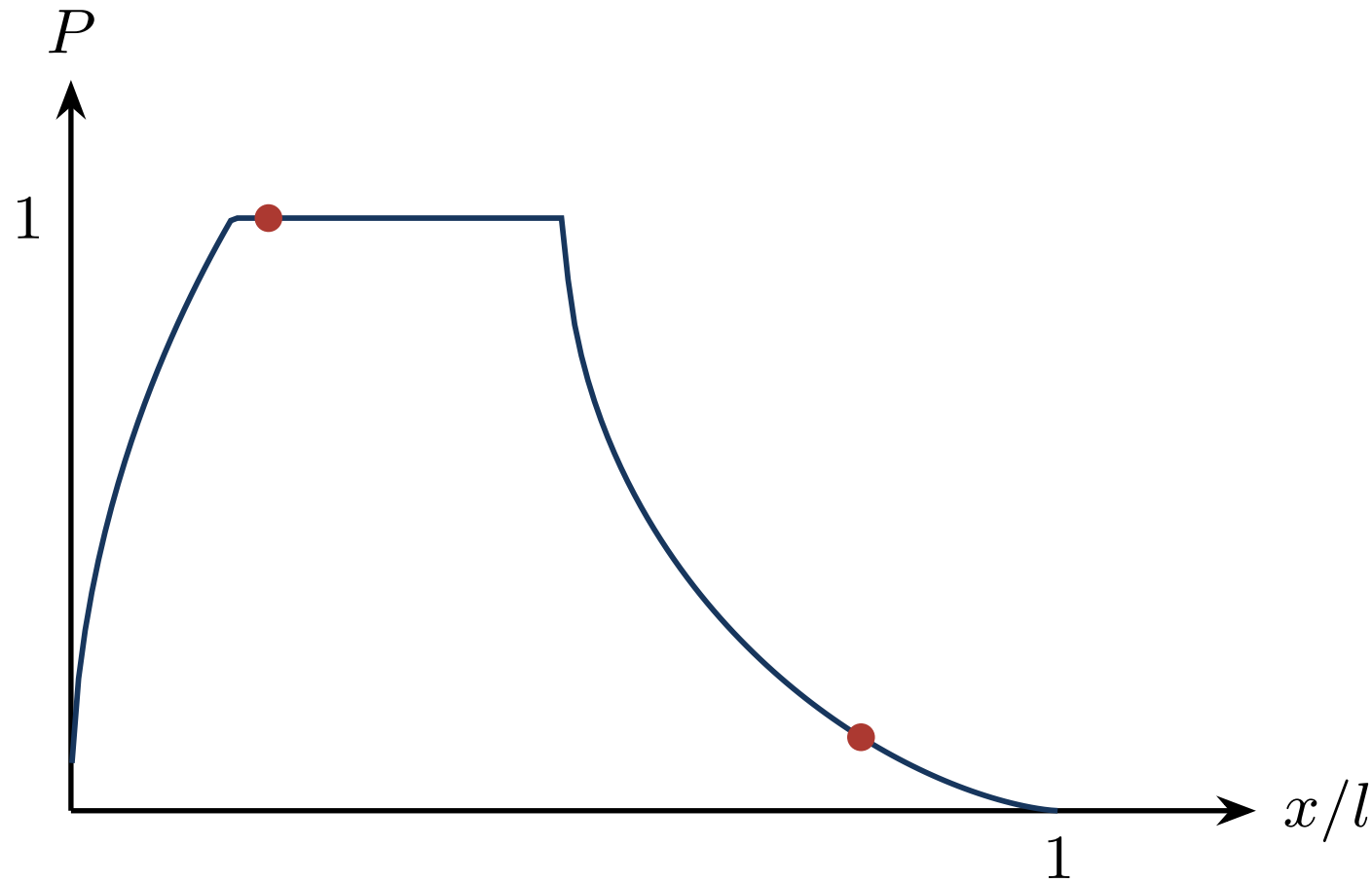


灰色运动线段；红色内部包络，蓝色端点候选轨迹

候选曲线需删去被其他时刻覆盖的弧



一个周期内的覆盖：左侧为全部，右侧宽度 $2A$



$P(0.2l) = 1$，$P(0.8l) = 0.124027$；按扫掠并集计算